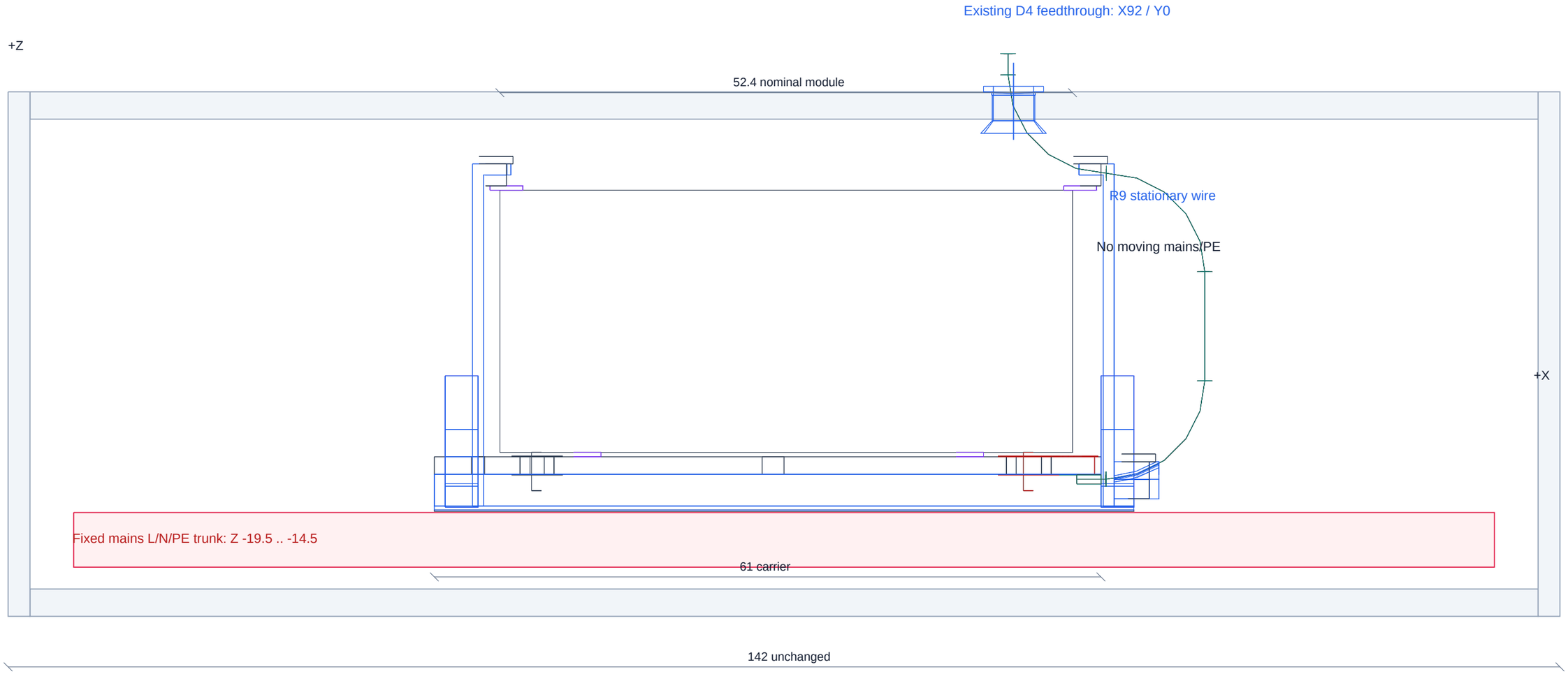
# Stationary module, carrier and finite secondary leadout

STATIONARY-R1 | mm | 2026-10-23 | unchanged core / trunk / endcaps | CONDITIONAL PROTOTYPE




Saved CAD: 54 individually valid solids; all-pair intersections independently enumerated.

Module nominal bottom/top -9.00 / +15.00; carrier bottom/top -11.00 / -9.40. Source pin length 2.5 .. 4.5, not a fixed 3.5. Min/max pins are separate editable source-lead-extremes.FCStd, never presented as a noncolliding nominal assembly.

No new hole or wall thinning. The port route ends at X91.5,Y +/-0.8,Z27.5, tangent +Z; collar FPC clamp remains an external contract. Wire loop outer X110.3. This section is a wireframe projection, not proof of assembly motion or sealed thermal performance.

# Lead engagement, tolerance envelope and carrier lands

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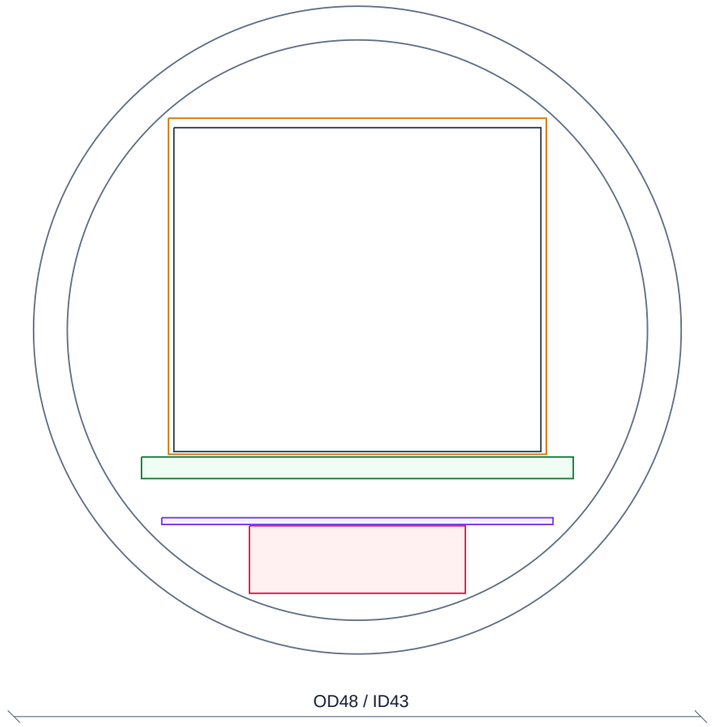
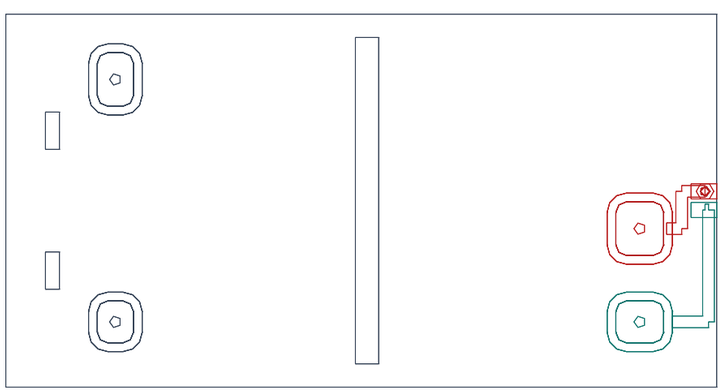
Y/Z section bounds (not an all-X silhouette)




Orange: source + process body envelope

Bore gap >= 0.465 (nominal bore only)



## Independent coupled vertical corners (32)

| | |
|---|---|
| PCB bottom | -11.00 +/-0.05 |
| PCB thickness | 1.60 +/-0.10 |
| Standoff | .40 +/-0.05 |
| Body height | 24.0 +/-0.5 source |
| Pin length | 2.5 .. 4.5 source |
| Min projection below Cu | .280 |
| Longest lead tip Z | -13.700 |
| Worst deck top Z | -13.850 |
| Lead / deck gap | .150 |
| Film / nominal trunk gap | .075 |
| Worst body top Z | +15.700 |

## Actual saved pin centres (X / Y), mm

| | |
|---|---|
| AC_N | 48.4 / -10.4 |
| AC_L | 48.4 / +10.4 |
| V_MINUS | 93.4 / -10.4 |
| V_PLUS | 93.4 / -2.4 |

Mechanical pin apertures include general drawing tolerances.

Nominal pins do NOT contact copper rings. Solder bridge / plated-aperture process UNMET. Primary branch wiring/fuse/connectors are not captured or released. The 2 x 28 slot is not a certified creepage barrier.

Deck material, 0.50 mm total tolerance allocation, core bore tolerance/ovality and source installation clearances remain open. Minimum lead projection is geometric engagement, not solder quality acceptance.

# Secondary sleeve, retention and assembly limitations

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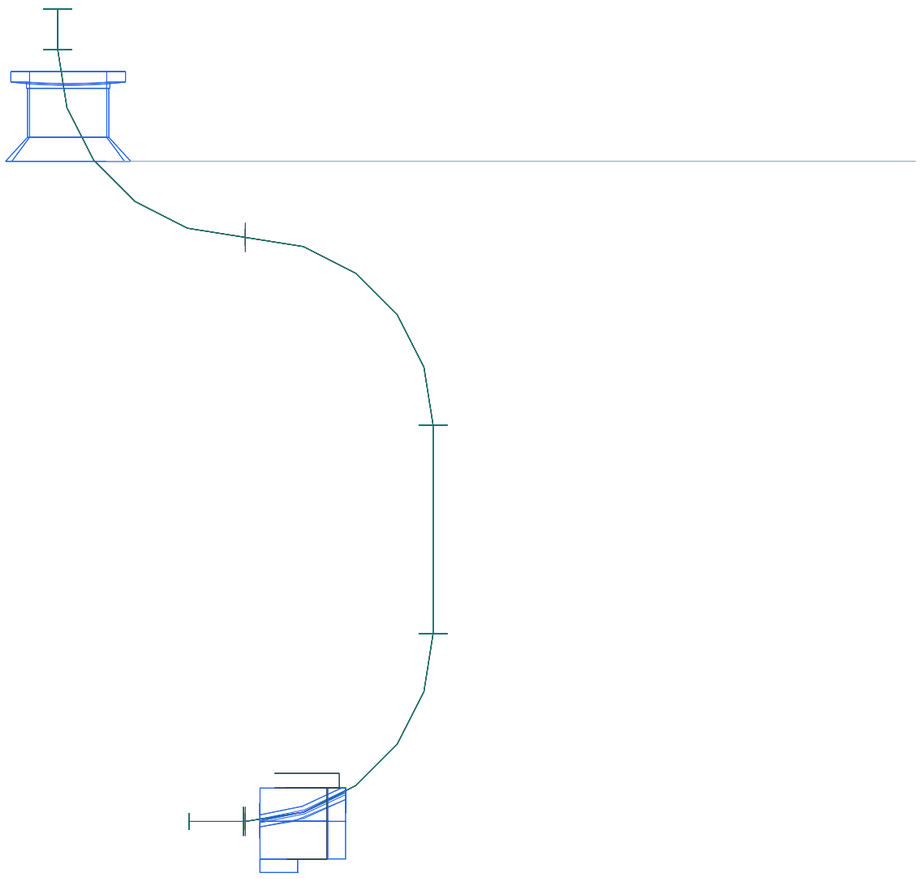
## Finite max-OD wire / split sleeve



Tensility30-02075: OD1.5 +/-0.10, solid20AWG

Source minimum bend R9; each formed length54.462

plus2.600 bare tail. No ampacity / fault I2t grant.

MEAN WELL generic PCB installation guide: 5 mm bottom insulation distance plus film, 10 mm around unit. These are not qualified by this tight layout. Module/body source approval is not barrel safety approval. Thermal/line derating in the sealed barrel, primary wiring, PE policy, variants and endcap clearances remain parent integration gates.

See DESIGN.md, checks.json, pair-checks.json and receipt.json for explicit assumptions and acceptance limits.

## Geometric screen is deliberately bounded

Wire / sleeve sampled minimum gap: 0.0112 mm.

Sleeve tube OD3.90 / ID3.70; proposed IDmin3.65.

Forming R9..9.1 and common XY placement +/-0.02 are

fixture allocations, not wire/sleeve supplier guarantees.

Sleeve split, wall0.10, burr coverage and retention UNMET.

Maximum jackets touch each other; no wire/core contact.

Adjustable jaws and bore shoes: no core penetration.

Thread/torque/friction/creep and pull strength UNMET.

## Required assembly sequence - motion NOT proven

1. Set board/module/jaws on sled outside tube.

2. Insert sled before forming the secondary loop.

3. Feed ends, form with end-access jig, fit split sleeve.

4. Complete clamp, inspect terminations, pull/isolation test.

5. Close parent-owned endcaps / primary harness design.

The verifier proves a preformed loop cannot slide axially into the intact core. Jig access, removal and forming loads are still UNMET; the completed CAD pose does not prove a buildable assembly sequence.